

RESTU ANGGORO KASIH

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Bachelor's degree in Mathematics with a strong foundation in data analysis, machine learning, and AI implementation. Experienced in managing projects involving data preprocessing, model development, hyperparameter tuning, performance evaluation, and requirement analysis. Proficient in statistical methods and programming, with a strong interest in applying data-driven approaches to problem-solving.

EDUCATION

YOGYAKARTA STATE UNIVERSITY

Bachelor of Science; Major in Mathematics

- Cum Laude with Cumulative GPA: 3.64/4.00

Sleman, IDN

September 2021 - January 2025

ORGANIZATION

Mathematics Student Association

2022

Staff, IT Division

- Performed maintenance and repairs on hardware inventory, including one computer, one router, and two printers, on a four-day rotation system to ensure optimal performance.
- Managed over 30 software installations and ensured the smooth operation of IT systems for all members.
- Researched, designed, and distributed 11 media publications in the form of videos and images related to the latest technological developments or interesting tech facts, coordinating with the communication and publication team.

Engineering and Technology Club (INFINITE)

2023

Staff, Competition Division

- Conducted research and monitoring of more than 15 IT-related competitions, gathering and distributing relevant information to members.
- Managed an archive of over 25 competition-related documents, including proposals and other documentation.

Student Executive Board

2023

Staff, Research, Data, and Digital Products

- Analyzed and visualized more than 10 survey datasets using Excel or Python to support strategic decision-making.
- Designed and produced over 10 informative infographics based on data analysis results, collaborated with the communication and publication team to ensure consistency and effectiveness in media content.

PROJECTS

Tea Leaf Disease Classification (Undergraduate Thesis)

- Developed an EfficientNetV2S-based image classification model utilizing transfer learning to identify and categorize six distinct tea leaf diseases, achieving 98.33% accuracy in disease detection on the validation dataset.
- Optimized model accuracy from 94.33% to 98.2% through hyperparameter tuning, data augmentation, and model architecture optimization.
- Applied advanced data preprocessing techniques (image normalization, contrast enhancement, and augmentation) on 1,500 images, improving feature extraction and model generalization.
- Evaluated model performance with 98.33% accuracy, 0.1169 loss, 98.33% precision, 98.36% recall, 98.33% F1-score, and AUC: 0.9976, ensuring high classification reliability.

Land Use Land Cover (LULC) Classification & Web Deployment (Internship Report)

- Designed an LULC classification system using Google Earth Engine (GEE) and Python, processing Sentinel-2 imagery with CNN, achieving 90.31% accuracy and Cohen's Kappa 85.49%.
- Built and deployed a Flask-based REST API and a React.js frontend, improving accessibility and scalability for geospatial data analysis.
- Enabled automatic area calculation for 4 land cover types (Water, Building, Plantation Forest, Herbaceous)

Customer Segmentation Using Birch Clustering

- Built a BIRCH clustering model to segment 2,000+ customers based on purchasing behavior and demographic data, improving customer targeting.
- Optimized clustering performance by selecting optimal clusters (Elbow Method), achieving strong metrics (Silhouette Score: 0.695, Davies-Bouldin Index: 0.440), and improving efficiency by 40% through feature selection, PCA, engineering, normalization, and data cleaning.
- Derived actionable insights by analyzing spending behavior, identifying 3 core customer segments, and informing marketing strategies to boost campaign effectiveness.

Scraping-Based Data Collection and Classification

- Implemented a web scraping system using Google Play Scraper and BeautifulSoup to extract 4,000+ user reviews for sentiment analysis, applying text preprocessing techniques such as regular expressions, tokenization, stopwords removal, stemming, and slang word normalization to improve data quality.
- Generated actionable insights from sentiment analysis by visualizing trends using WordCloud and Seaborn, identifying 3 dominant sentiment categories that influenced model interpretation.
- Performed sentiment classification using multiple machine learning models, applied SMOTE to address class imbalance, performed hyperparameter tuning, and achieved 96.92% accuracy using SVM.

Bank Fraud Transaction

- Conducted EDA and feature engineering on 2,512 transactions (Jan 2023–Jan 2024), achieving high-quality dataset.
- Built an ensemble of 7 unsupervised models (e.g., Isolation Forest, LOF, GMM) to detect 59 anomalies (2.35%) using majority voting (≥ 4 models); validated by domain fraud rules.
- Derived key insights: anomalies showed 115% higher amounts, impossible travel ($>2,000$ km/h), peak fraud at 17–18h, and high anomaly rates in debit/branch channels.

Credit Card Report Dashboard (MySkill Course Project)

- Developed dashboard using Google Looker Studio for credit card performance metrics visualization
- Analyzed and emerged key data insights from 3 core business areas (transaction patterns, delinquent account risk profiles, and revenue performance) and 3 key segments (Customer Demographics, Revenue Profile, and Risk Profile) to support strategic business decisions.

AWARDS

- **3rd Place** – LIKMI, Applied Technology Business Category, Yogyakarta State University 2023
Project: *“FO-Shion (Find Your Shade Cushion): A Spark AR Studio-Based Solution for Choosing the Right Cushion Shade for Skin Tone”*.
- **Finalist** – TrackAML Hackathon PPAK 2023
Project: *“A Combined Fraud Detection and Ensemble Learning System for AML and Terrorism Indicators in Anomaly Data Analysis”*.
- **Finalist** – Abdidaya Ormawa, PPK Ormawa, Ministry of Education, Culture, Research, and Technology 2023
Project: Community service to build a climate-resilient village through agroforestry, biogas, biopore infiltration systems, and MSME empowerment using agroforestry products like eco-printing.

ADDITIONAL

Technical Skills: Python, SQL, Tableau, MATLAB, JavaScript, R, HTML/CSS, Machine Learning, Deep Learning, Data Visualization, Web Scraping, Microsoft Office, Flask API, AWS, React JS, Node JS, Big Data, RESTful API, Power BI, Google Looker Studio.

Languages: Native in Indonesian; Conversational Proficiency in English.

Certifications & Training:

- Data Analyst with Python – IBM (Coursera), 2024
- Fundamentals of AI, Data Visualization, SQL, Python, Web Programming, JavaScript, Front-End Web, AWS, Back-End – Dicoding, 2024 & 2025.
- Google Looker Studio – MySkill, 2025